Synechron L2 – Technical Deep-Dive & Banking Scenarios

Synechron L2 • Senior DevOps / Platform Engineering / SRE / Banking • Interview prep

A. High-Probability L2 Deep Dive

Q: Platform engineering vs DevOps?

Spoken answer: DevOps is a broad culture and operating model joining development and operations. Platform engineering focuses on internal platforms and reusable capabilities that create safe developer golden paths. There is significant overlap in CI/CD, IaC, Kubernetes, observability and security.

Q: How would you build a self-service platform?

Spoken answer: Start with standard templates for service creation, CI/CD, infrastructure, observability and security. Provide approved Terraform modules and deployment templates, central secrets, RBAC and auditability. Measure adoption, lead time, deployment failure and incidents.

Q: How do you secure CI/CD?

Spoken answer: Protected branches, PR reviews, least-privilege identities, secret management, SAST, dependency and image scanning, artifact integrity, controlled environments and production approvals. Prefer short-lived cloud identities such as OIDC where supported.

Q: What is IRSA?

Spoken answer: IAM Roles for Service Accounts maps a Kubernetes service account to an AWS IAM role so pods can obtain AWS permissions without long-lived access keys. It supports least privilege. EKS also provides Pod Identity as a newer workload-identity mechanism.

Q: ALB vs NLB?

Spoken answer: ALB is Layer 7 and supports HTTP/HTTPS host and path routing. NLB is Layer 4 for high-performance TCP/TLS/UDP and static IP use cases. The choice depends on protocol and routing needs.

Q: HPA vs VPA?

Spoken answer: HPA changes replica count using metrics. VPA adjusts pod resource requests/limits. HPA is common for horizontal scaling; VPA requires care because resource changes can restart pods depending on mode.

Q: Requests vs limits?

Spoken answer: Requests influence scheduling and represent expected resource need. Limits cap consumption. Correct sizing improves scheduling and stability; poor limits can cause throttling or OOMKilled.

Q: What is a PDB?

Spoken answer: A PodDisruptionBudget limits voluntary disruptions during maintenance such as node draining, helping preserve availability. It does not protect against every failure.

B. GitOps / Jenkins / Secrets

Q: Jenkins controller vs agent?

Spoken answer: The controller orchestrates jobs and agents execute steps. I prefer isolated or ephemeral agents where appropriate, declarative pipelines and code-reviewed pipeline definitions.

Q: What is GitOps?

Spoken answer: Git is the desired-state source of truth. A controller such as Argo CD reconciles the cluster to Git, providing traceability, repeatability and auditable deployment history.

Q: Argo CD OutOfSync?

Spoken answer: I inspect the desired/live diff, sync status and events, then determine whether drift is expected, generated by another controller or caused by manual change. I fix the source of truth rather than blindly forcing sync.

Q: Kubernetes secrets?

Spoken answer: I avoid plaintext secrets in Git. I use an approved managed secret store/CSI integration or equivalent, with RBAC, encryption, rotation and auditability.

C. Linux / Python / Automation

Q: High CPU on Linux?

Spoken answer: Confirm host-wide versus process-specific impact with top/ps, inspect load and CPU breakdown, check recent changes and logs, identify the process and mitigate according to impact, then validate and perform RCA.

Q: Disk 95% full?

Spoken answer: Use df and du, inspect logs, temporary files, container layers and deleted-open files. Do not blindly delete. Rotate/archive safely, remove approved temporary data and fix the root cause such as missing log rotation.

Q: Python in SRE?

Spoken answer: I use Python for API automation, health checks, data processing and operational tooling. I make scripts idempotent, observable, configurable and safe to rerun.

Q: Production-safe automation?

Spoken answer: Define inputs and failure conditions, make it idempotent where possible, validate, use least privilege, log outcomes, handle retries carefully and provide rollback or compensating actions. Version and review it like application code.

D. Banking Scenario Drills

Q: Critical release but business approval missing?

Spoken answer: I state that the release is technically ready but approval is outstanding. I do not represent it as approved. I contact the accountable approver, update the change record and escalate according to process if the window is at risk.

Q: Database migration cannot be rolled back?

Spoken answer: Use backward-compatible expand-and-contract patterns where possible. Establish recovery points, deploy application changes compatible with both schemas, validate, and only remove old schema elements after confidence. Rollback becomes application rollback plus safe recovery or forward-fix.

Q: Latency rises but errors are normal after release.

Spoken answer: Compare percentile latency before/after, inspect traces, downstream dependencies, DB queries, CPU/memory, network and traffic. If the release is implicated and SLO impact is significant, consider rollback or traffic reduction.

Q: Critical security vulnerability on release day?

Spoken answer: Verify true positive and whether the vulnerable component is deployed/reachable. Assess exploitability with Security, then remediate/upgrade or use a formally approved exception if immediate remediation is impossible. Document owner and remediation date.